

Where Will the Swarm Swarm: Predicting Deployments under Lloyd Dynamics

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Abstract—We predict the final deployment of unmanned aircraft systems (UAS) executing a specified centroid-based law driven by an unknown spatial demand density. We use the positions and velocities of all UAS at different snapshots in time to estimate their final deployment. We show that a single snapshot can be consistent with distinct demand densities and different attracting deployments, and that the worst-case density estimation risk under Kullback–Leibler loss is infinite. Near a local minimum of the coverage objective, however, the largest distance between any vehicle’s current position and its eventual position can be bounded within a constant multiple of the maximum observed speed. Under additional local regularity, a Newton-type correction yields a prediction error that is quadratic in speed, providing a sharper bound near convergence. We numerically estimate the unknown demand density using a positive radial-basis model fitted to cumulative motion constraints. Simulations with different demand densities evaluate prediction accuracy and computational cost under known and estimated controller gains.

Index Terms—UAS swarms, UAV deployment, trajectory prediction, Lloyd dynamics, inverse coverage.

I. INTRODUCTION

Coverage control is a widely used approach to coordinating swarms of unmanned aircraft systems (UAS) [1]. A standard class of coverage controllers is based on a continuous-time version of Lloyd’s algorithm [2]. Lloyd’s algorithm alternates between partitioning the domain into Voronoi cells and moving each vehicle to the demand-weighted centroid of its cell. In the continuous-time version studied here, each vehicle moves with velocity proportional to the displacement from its current position to that centroid, while the cells and centroids evolve with the swarm configuration. This motion distributes the vehicles according to a spatial demand density. DARPA’s OFFensive Swarm-Enabled Tactics (OFFSET) program, which investigated large air/ground teams for urban operations [3], illustrates the practical interest in coordinated swarms. Such applications motivate anticipating where a swarm will concentrate from observations of its motion. We study this inverse question for vehicles following the specified coverage law, which we call normalized continuous Lloyd dynamics, when the spatial demand density is unknown.

Recent UAV research develops decentralized goal-convergent planning [4], sensing/communication-aware coordination [5], and learned swarm control [6]. Quantization-based deployment design connects spatial demand to wireless UAV placement [7]. Adaptive coverage learns fields from direct measurements [8],

including noisy spatial observations [9], time-varying processes [10], modified coverage objectives [11], and learned perception, communication, and action [12]. Our observer instead receives positions and velocities from a fixed controller and no samples of the hidden field.

Coverage path planning determines routes for vehicles to inspect or sense an area [13], [14]. Our goal is instead to predict where vehicles will eventually settle under a given coverage control law. The coverage framework can encode diverse spatial priorities through the demand density: localized peaks can represent targets or locations of interest, while broader regions of elevated density can represent areas requiring distributed coverage. Lloyd dynamics translate these priorities into coordinated motion toward an eventual deployment. Recent inverse methods infer swarm interaction topology and finite-horizon trajectories [15], or intent classes from aerial observations [16]–[18]. Other work infers controller goals and gains [19] or movement costs [20]. Here the latent objective is a common spatial density and the prediction target is the trajectory’s limiting configuration.

The rest of the paper presents the model (Section II), theoretical results (Section III), and prediction methods (Section IV), followed by numerical experiments and conclusions. The appendices, which are provided in [21], contain proofs and details of the constructions.

II. OBSERVED MOTION AND THE PREDICTION OBJECTIVE

Let $\Omega \subset \mathbb{R}^k$ be bounded and convex with nonempty interior. A fixed density f satisfies $f \geq 0$ and $\int_{\Omega} f = 1$. For distinct vehicle positions $X = (x_1, \dots, x_n)$, define their Voronoi cells $V_i(X) = \{z \in \Omega : \|z - x_i\| \leq \|z - x_j\| \forall j\}$, masses $M_i = \int_{V_i} f$, and weighted centroids $c_i = M_i^{-1} \int_{V_i} z f$. Write $C_f = (c_i)_{i=1}^n$. For known gain $\gamma > 0$, the normalized Lloyd law is

$$\dot{X} = F_f(X) := \gamma(C_f(X) - X). \quad (1)$$

At a forecast time t_0 , the complete positions $X_0 = X(t_0)$ and vector velocities $V_0 = \dot{X}(t_0)$ give

$$C_0 = X_0 + V_0/\gamma, \quad \int_{V_i(X_0)} (z - c_i) f(z) dz = 0. \quad (2)$$

The analysis and reconstruction below assume that the constant gain γ is known, for example from controller settings or calibration. We assume a known gain for the analysis below and address its estimation later in the paper.

With $M_f = \text{diag}(M_i I_k)$, define

$$\mathcal{H}_f(X) = \sum_i \int_{V_i(X)} \|z - x_i\|^2 f(z) dz. \quad (3)$$

Then, we have $\nabla \mathcal{H}_f = 2M_f(X - C_f)$, and $F_f = -\frac{\gamma}{2} M_f^{-1} \nabla \mathcal{H}_f$ [1], [2]. To avoid technicalities and focus on local prediction, we adopt the following standard assumptions, consistent with the convergence analysis of Lloyd dynamics [1].

Assumption 1. *The density f is smooth and strictly positive. The trajectory $X(t)$ has pairwise distinct vehicle positions and converges to a nondegenerate local minimum X_* of the coverage objective, meaning $\nabla \mathcal{H}_f(X_*) = 0$ and $H_* := \nabla^2 \mathcal{H}_f(X_*) \succ 0$. The limiting sites are distinct and lie strictly inside Ω . The centroid map is continuously differentiable near the trajectory and its limit.*

We seek to predict each vehicle's eventual position, without assuming that the resulting deployment is globally optimal. For $Z = (z_i)_i$, set $\|Z\|_b = \max_i \|z_i\|_2$; otherwise norms are Euclidean and matrix norms are induced operator norms. Two simple endpoint predictions are the current positions X_0 and the current centroids $C_0 = X_0 + V_0/\gamma$. The first assumes no further motion. The second assumes that each vehicle approaches its current centroid, ignoring how that centroid changes as the swarm moves and the Voronoi cells evolve.

III. IMPOSSIBILITY AND LOCAL PREDICTION GUARANTEES

What are the fundamental limits of reconstructing the demand density and predicting the final deployment from a single snapshot of the swarm's positions and velocities? This section addresses this question from a theoretical perspective. We first show that a single snapshot can be consistent with distinct demand densities and different attracting deployments, so neither quantity is uniquely determined in general. We then establish local prediction guarantees.

For a fixed observed snapshot (X_0, V_0) , let $\mathcal{F}$ be the set of all normalized densities satisfying Assumption 1 and the centroid constraints in (2). Every density in $\mathcal{F}$ therefore produces the same observed snapshot. To quantify the remaining reconstruction uncertainty, define the KL loss $D_{\text{KL}}(f \| q) = \int_{\Omega} f(z) \log \frac{f(z)}{q(z)} dz$ and the minimax risk

$$R^* = \inf_{\hat{f}} \sup_{f \in \mathcal{F}} \mathbb{E} D_{\text{KL}}(f \| \hat{f}). \quad (4)$$

Here, $\hat{f}$ is a density estimate based only on the observed snapshot, and the expectation allows for randomized estimates. The infimum selects the best reconstruction rule, while the supremum considers the most unfavorable density compatible with the observations. The following result shows that this worst-case error is unbounded.

Theorem 1. *We have $R^* = \infty$. Moreover, a single snapshot need not uniquely determine the final deployment.*

Proof sketch. We construct a family of smooth, positive densities that satisfy the same centroid conditions and therefore produce the same observed snapshot. Within this family, we concentrate a fixed amount of mass into progressively smaller

regions while preserving these conditions and the convergence assumptions. Since the observations cannot distinguish these densities, any reconstruction rule incurs arbitrarily large KL loss for some member of the family. Finally, a two-vehicle example shows that two densities producing the same snapshot can lead to different attracting final deployments. $\square$

This impossibility result still leaves room for useful prediction. Theorem 1 concerns worst-case single-snapshot inference; infinite KL risk allows arbitrarily concentrated fields without common derivative bounds. It does not imply infinite position error for the endpoints. Practical prediction can exploit a better-behaved local regime. In fact, along a fixed trajectory approaching a nondegenerate minimum, positive curvature links remaining displacement to speed.

Theorem 2. *There is a neighborhood U of X_* and constants $\kappa_1, \kappa_2 < \infty$ such that whenever $X_0 \in U$,*

$$\|X_* - X_0\|_b \leq \frac{\kappa_1}{\gamma} \|V_0\|_b, \quad \|X_* - C_0\|_b \leq \frac{\kappa_2}{\gamma} \|V_0\|_b, \quad (5)$$

Both bounds apply at all sufficiently late times.

Near convergence, slower motion gives a smaller prediction error bound, although the constants depend on the density and swarm. We next show how accounting for changes in the velocity field improves the error bound from linear to quadratic in speed. To retain how neighboring vehicles change the velocity field, freeze its Jacobian $J_0 = DF_f(X_0)$, where D represents the Jacobian operator, and solve

$$Y_{\infty} = X_0 - J_0^{-1} V_0. \quad (6)$$

This is a Newton step for $F_f(X) = 0$. It is also the endpoint of $\dot{Y} = V_0 + J_0(Y - X_0)$, $Y(0) = X_0$, when J_0 is Hurwitz (all eigenvalues have negative real parts). Replacing J_0 by $-\gamma I$ gives the current-centroid predictor.

Theorem 3. *Suppose DF_f is locally Lipschitz near X_* . There are $T, K < \infty$ such that $\forall t_0 \geq T$, J_0 is Hurwitz, invertible and*

$$\|X_* - Y_{\infty}\| \leq K \|V_0\|^2. \quad (7)$$

The bounds for predictions based on the current positions and centroids are proportional to the observed speed. Using the exact current Jacobian improves this bound to one proportional to the square of the speed. To implement this correction, we need to estimate the Jacobian. In the next section, we reconstruct the unknown demand density from observed motion and compute the Jacobian of the resulting velocity model. The prediction accuracy then also depends on how closely this estimated Jacobian matches the true one.

IV. LEARNING AND PREDICTING FROM MOTION

To predict the final deployment, we first estimate the unknown demand density from the swarm's observed positions and velocities. We combine the centroid constraints from successive snapshots to fit a single density. We then use this estimate either to simulate Lloyd dynamics or to compute the Jacobian correction.

Algorithm 1 Density reconstruction from UAS motion

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1: Normalize the basis on  $\Omega$ ; initialize empty  $A, a$ .
2: for each observed time  $s \leq t_0$  do
3:   for each observed UAS  $i$  do
4:     Set  $r = (s, i)$  and  $V_r = V_i(X^{(s)})$ .
5:     Set  $b_r = x_i^{(s)} + v_i^{(s)}/\gamma$  and  $q_r = |V_r|/|\Omega|$ .
6:     Integrate (9) on  $V_r$  and append  $k$  rows to  $A, a$ .
7:   end for
8: end for
9: Solve the convex simplex-constrained problem (10) at  $t = t_0$ .
10: return  $\hat{f}_{t_0} = g_{\hat{w}_{t_0}}$  from (8).

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A. Density Reconstruction using Radial Basis Functions

We model the density using a fixed collection of Gaussian basis functions $\phi_j(z) = Z_j^{-1} \exp[-\|z - \mu_j\|^2/(2\sigma_j^2)]$, $j = 1, \dots, p$ normalized over Ω . Here, $Z_j = \int_{\Omega} \exp[-\|z - \mu_j\|^2/(2\sigma_j^2)] dz$ ensures that $\int_{\Omega} \phi_j(z) dz = 1$. In practice, for $\Omega = [0, 1]^k$, we place the centers μ_j on a Cartesian midpoint grid with m points per coordinate, giving $p = m^k$ Gaussian basis functions. Let

$$g_w(z) = \frac{\eta}{|\Omega|} + (1 - \eta) \sum_{j=1}^p w_j \phi_j(z), \quad w \in \Delta_p, \quad (8)$$

where $\Delta_p = \{w \geq 0 : \mathbf{1}^\top w = 1\}$ and $0 < \eta < 1$. For each observed vehicle/time pair r , its measured centroid $b_r = x_r + v_r/\gamma$ and cell V_r impose $\int_{V_r} (z - b_r) g_w(z) dz = 0$. Set $q_r = |V_r|/|\Omega|$. Dividing these moments by q_r and stacking coordinates produces $Aw + a = 0$, with rows

$$\begin{aligned} A_{(r,d),j} &= \frac{1-\eta}{q_r} \int_{V_r} (z_d - b_{r,d}) \phi_j(z) dz, \\ a_{(r,d)} &= \frac{\eta}{q_r |\Omega|} \int_{V_r} (z_d - b_{r,d}) dz. \end{aligned} \quad (9)$$

With N_t scalar rows available through time t and $\lambda > 0$, solve

$$\hat{w}_t = \arg \min_{w \in \Delta_p} \left\{ \frac{\|Aw + a\|_2^2}{2N_t} + \frac{\lambda}{2} \|w - \mathbf{1}/p\|_2^2 \right\}. \quad (10)$$

Algorithm 1 provides the full reconstruction procedure, including how we update the density weights whenever a new snapshot arrives, using constraints from all snapshots observed so far. Each constraint retains the Voronoi cell from the time its observation was made. The fit need not reproduce every observed centroid exactly.

Our density model uses Gaussian components with fixed positive widths and nonnegative weights summing to one. These choices impose a common upper bound on every fitted density, preventing the model from pathologically placing a fixed amount of probability mass into arbitrarily small regions, as in the construction of Theorem 1.

B. Jacobian Estimation and Endpoint Prediction

One trajectory need not span all nk configuration directions needed to estimate a full Jacobian from successive differences. Instead, we probe the fitted field at $X_0 \pm h_\ell e_\ell$, where e_ℓ is the ℓ th standard basis vector in $\mathbb{R}^{nk}$, and $h_\ell > 0$ is a small perturbation step chosen so that both $X_0 \pm h_\ell e_\ell$ have distinct vehicle positions inside Ω . For each perturbation, we rebuild the Voronoi cells. Column ℓ of the estimated Jacobian is then

$$\hat{J}_{:\ell} = \frac{1}{2h_\ell} (F_{\hat{f}_t}(X_0 + h_\ell e_\ell) - F_{\hat{f}_t}(X_0 - h_\ell e_\ell)) \quad (11)$$

Algorithm 2 Model-probe Jacobian and anchored endpoint prediction

Require: Fitted density $\hat{f}_{t_0}$ from Algorithm 1; current interior configuration X_0 and measured velocity V_0 .

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1: for  $\ell = 1, \dots, nk$  do
2:   Choose  $h_\ell \in (0, h]$  so both probes have distinct sites in  $\text{int } \Omega$ .
3:   for  $\sigma \in \{-1, +1\}$  do
4:     Set  $X^\sigma = X_0 + \sigma h_\ell e_\ell$  and rebuild its cells  $V_i^\sigma$ . Compute
       the corresponding centroids  $\hat{c}_i^\sigma$  using the density  $\hat{f}_{t_0}$ .
5:     Set  $\hat{F}^\sigma = \gamma((\hat{c}_i^\sigma)_{i=1}^n - X^\sigma)$ .
6:   end for
7:   Set  $\hat{J}_{:\ell} = (\hat{F}^{+1} - \hat{F}^{-1})/(2h_\ell)$ .
8: end for
9: Set  $s = \max \text{Re } \lambda(\hat{J})$  and flag ( $s < -\delta$ ).
10: if  $\hat{J}$  is singular then
11:   return the flag and “no unique correction.”
12: end if
13: Solve  $\hat{J}d = -V_0$  using the current measurement.
14: return  $X_0 + d$ ,  $\hat{J}$ , and the stability flag.

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Algorithm 2 provides the entire procedure. The prediction solves $\hat{J}d = -V_0$ and returns $X_0 + d$, using the current measured velocity even when $F_{\hat{f}_t}(X_0) \neq V_0$. This can reduce endpoint bias from a fitted field that does not reproduce the current velocity. An alternative that we shall investigate numerically is what we call the “full simulation” scenario, where we instead starts from the currently observed positions X_0 and run Lloyd dynamics using the fitted density $\hat{f}_{t_0}$.

If $\hat{J}$ is singular, Algorithm 2 reports that no unique Jacobian-based prediction is available. Otherwise, it returns $X_0 - \hat{J}^{-1}V_0$, flagging the prediction if the stability check fails.

To assess the benefit of recomputing the Jacobian at each forecast, we also evaluate a reuse baseline. This baseline stores a stable Jacobian $\bar{J} = \hat{J}(t_r)$ at time t_r . Later predictions use this same matrix with newly measured positions and velocities via $\hat{X}_{\text{reuse}}(t) = X(t) - \bar{J}^{-1}\dot{X}(t)$, $t \geq t_r$.

V. NUMERICAL EXPERIMENTS**A. An Eight-Peak Deployment**

Thirty UAS follow (1) on $[0, 1]^2$ with $\gamma = 1$, starting around $(0.5, 0.5)$. The synthetic target density is concentrated to 8 locations, simulating the scenario where the UAS autonomously converge to the 8 targets. Precise description of the setup is provided in Appendix D. The snapshots are obtained at times $\{0, 0.1, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 640, 768, 1024\}$ and one at a time, for a total count of 17 snapshots. Closely spaced early samples resolve the transient and geometric spacing covers the longer evolution. The estimated Jacobian was stable for all times $t \geq 512$.

Fig. 1 shows how the cumulative reconstruction improves as the swarm moves and its cells change. At $t = 0$, the single snapshot leaves substantial ambiguity; by $t = 4$, the estimate resolves all eight peaks. In Fig. 2, we provide the root mean squared Euclidean error over labeled UAS against the true limiting deployment. Current centroid estimates are provided by (2). The reuse baseline stores the first stable forecast matrix at $t_r = 512$. We observe that that full simulation is more

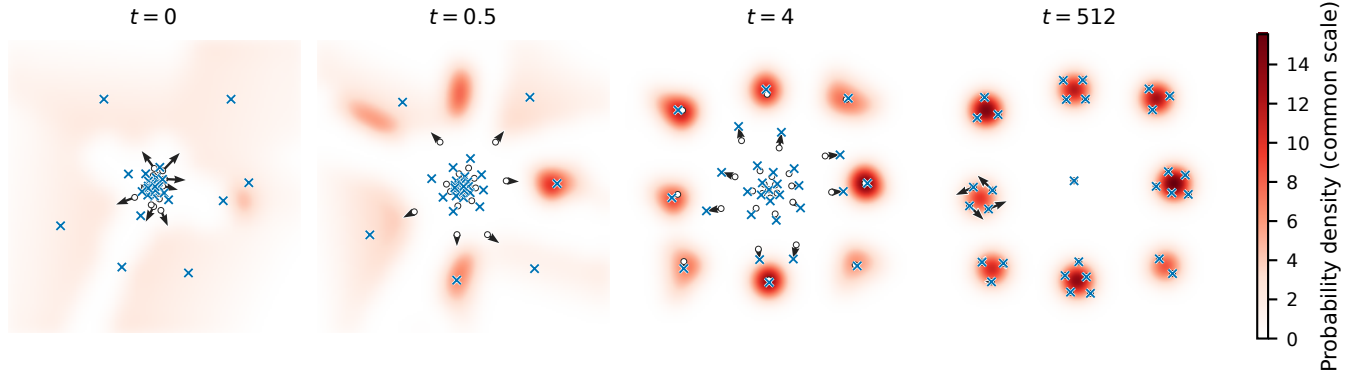


Fig. 1. Successive cumulative density estimates at $t = 0, 0.5, 4$, and 512 for the eight-peak scenario. All panels use the same white-to-red density scale. White circles ($\circ$) mark observed positions, dark arrows show measured velocities, and blue crosses ($\times$) mark current centroids $C = X + V/\gamma$. Arrow length multipliers are $0.25, 0.25, 1$, and 10^3 , respectively.

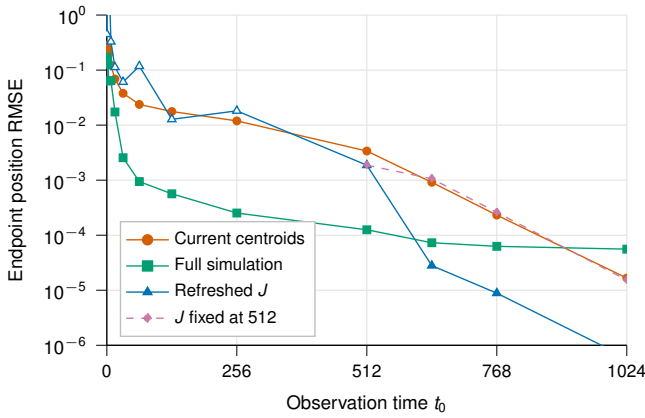


Fig. 2. Endpoint forecasts for the eight-peak scenario.

accurate at early times, while refreshing the Jacobian improves late predictions.

On an Apple M1, the refreshed prediction at 1024 takes 1.88 s (median of three runs), including probes, eigenvalues, and the solve; full simulation takes 41.1 s in one run. Using the previously computed LU factors of the reused Jacobian, solving for the endpoint correction takes 8.33 microseconds. This timing excludes the earlier Jacobian construction and factorization. These timings exclude fitting the density, which both methods require.

B. An A-Shaped Field with an Estimated Gain

We next use 24 UAS, initialized uniformly at random in $[0.1, 0.2] \times [0, 1]$. The demand is an “A shape” centered at $(0.75, 0.5)$ in the right half of the square. Here we assume that the gain γ is initially unknown. We use the profiling algorithm in Appendix E to jointly estimate γ and the demand density from cumulative motion observations.

Fig. 3 shows that the reconstructed field resolves the A around $t = 8$, while some UAS continue serving the positive background density outside it. The initial gain guess 1.2 becomes 0.8045 at the first update, $t = 0.5$, and 0.7958 at $t = 1024$, an eventual relative gain error of 0.53%. The

estimates fluctuate rather than improve monotonically; basis approximation and regularization can leave a small gain bias. As a separate control, holding the initial gain at 1.2 gives density KL 0.995 at $t = 1024$, compared with 0.00678 after gain estimation. The supplement reports the known-gain control, all eigenvalue checks, and refinement tests.

The fitted Jacobian is Hurwitz for $t \geq 512$. The reuse matrix is thus frozen at $t_r = 512$. Reusing the matrix from 512 gave nearly the same late accuracy as refreshing in this experiment, as shown partially in Fig. 4 (We omitted plotting the points $t > 640$ as the RMSE becomes very low.). The general trend of different methods is the same as in Fig. 2. While full simulation provides the best performance at early times, it is outperformed by centroid prediction later. In addition, Jacobian estimation provides the best performance eventually, due to the quadratic approach provided by the exact Jacobian (Theorem 3).

We also provide the KL risk of the fitted estimate for the two scenarios described above in Fig. 5, which shows that our density estimate is accurate for both cases. Finally, a three-dimensional density example is considered in Appendix F.

VI. CONCLUSION

We proposed algorithms to estimate the demand density and predict the final deployment of vehicle swarms from their observed motion under Lloyd dynamics. Full simulation using the estimated density performed well at early times, while the Jacobian approach was more accurate at later times and reduced computation through matrix reuse. Extensions to other swarm controllers remain future work.

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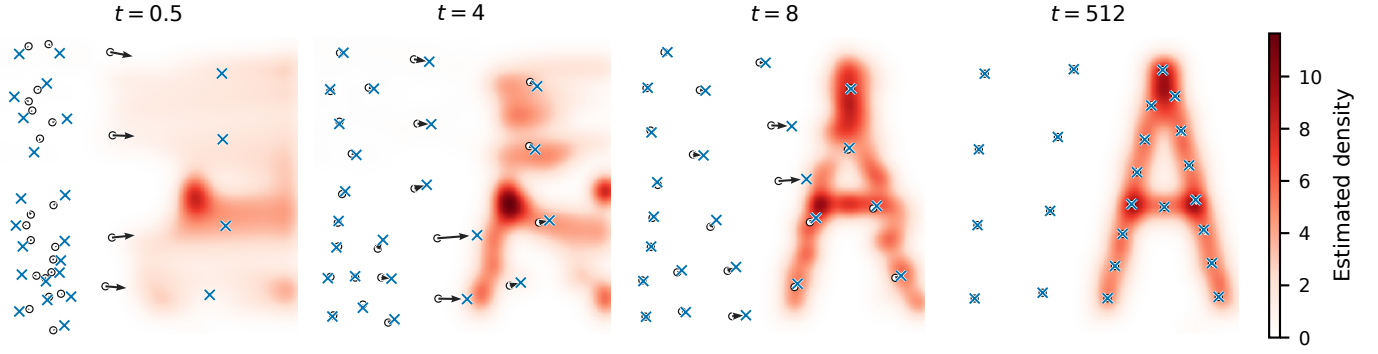


Fig. 3. Cumulative reconstruction of the A-shaped demand with an estimated gain at $t = 0.5, 4, 8$, and 512 . White circles (o) mark observed positions, dark arrows show measured velocities, and blue crosses (x) mark centroid estimates $\hat{X} + V/\hat{\gamma}_t$. Arrow multipliers are $0.25, 1, 1$, and 10^4 , respectively, and affect only the displayed velocities; centroid estimates are not magnified.

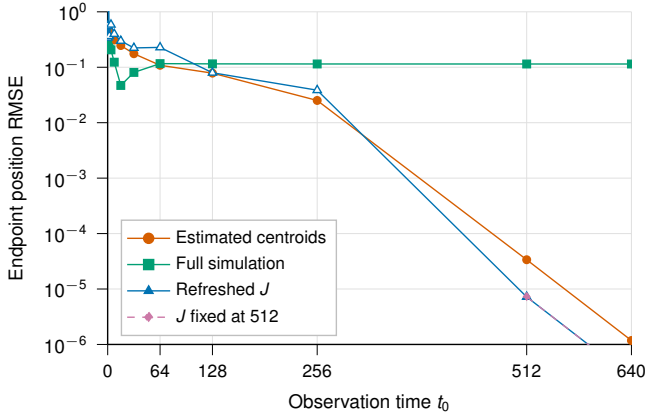


Fig. 4. Endpoint forecasts for the A-shaped demand.

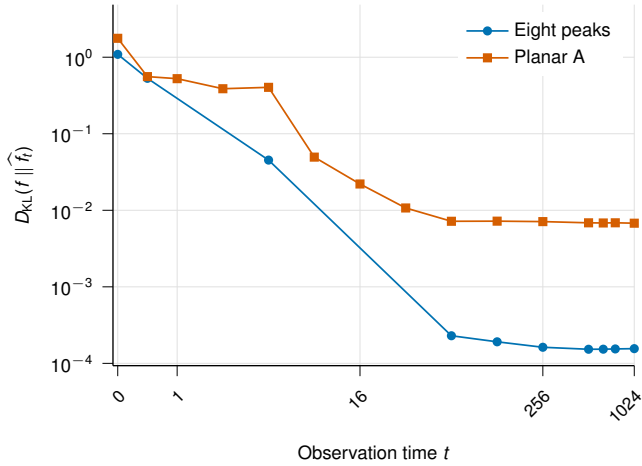


Fig. 5. KL risks for two scenarios.

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APPENDIX A – PROOF OF THEOREM 1

Let $\text{TV}(f, g) = \frac{1}{2} \int_{\Omega} |f - g|$ and $\text{JS}(f, g) = \frac{1}{2} [D_{\text{KL}}(f \| m) + D_{\text{KL}}(g \| m)]$, where $m = (f + g)/2$.

Moment-preserving perturbations. Choose a ball B compactly contained in one observed cell and a nonzero $\varphi \in C_c^\infty(B)$. The function $h = \partial_1^2 \varphi$ is nonzero: otherwise compact support would force φ , affine along each coordinate line, to vanish. For $p = 1, z_1, \dots, z_k$, two integrations by parts give

$$\int_B p(z) h(z) dz = \int_B \varphi(z) \partial_1^2 p(z) dz = 0.$$

Thus $f_{\pm} = f_0 \pm \varepsilon h$ are normalized, strictly positive for small ε , and have identical cell masses and centroids at X_0 . Disjoint sub-balls give infinitely many independent directions. The nondegenerate attracting minimum of f_0 persists under small smooth perturbations. On a finite trajectory segment leading into its attracting neighborhood, continuous dependence of solutions keeps the perturbed trajectory close to the baseline. It therefore enters the continued basin, preserving Assumption 1.

Positive KL risk. For $m = (f_+ + f_-)/2$ and every density q ,

$$\frac{1}{2} \sum_{\pm} D_{\text{KL}}(f_{\pm} \| q) = \text{JS}(f_+, f_-) + D_{\text{KL}}(m \| q).$$

The observation laws, hence the output laws of any randomized estimator, coincide for the two fields. Taking expectations and using Pinsker's inequality with $\text{TV}(f_{\pm}, m) = \text{TV}(f_+, f_-)/2$ gives $R^* \geq \text{JS}(f_+, f_-) \geq \text{TV}(f_+, f_-)^2/2 > 0$. This also covers a fixed noisy channel depending only on snapshot velocity.

Infinite KL risk. Fix a baseline f_0 , initial cell masses m_i , centroids c_i , and limiting minimum X_*^0 . Inside each initial cell, choose a simplex containing c_i in its interior, with vertices a_{ij} away from the cell interfaces at X_*^0 . Such vertices exist because those interfaces form a finite union of hyperplanes. Choose positive weights with $\sum_j w_{ij} = 1$ and $\sum_j w_{ij} a_{ij} = c_i$. Let $\psi_{ij,r}$ be a normalized radial smooth bump of radius r centered at a_{ij} , and set

$$p_r = \sum_{i,j} m_i w_{ij} \psi_{ij,r}, \quad f_r = (1 - \varepsilon) f_0 + \varepsilon p_r.$$

For small r and $0 < \varepsilon \leq 1/2$, these smooth densities satisfy $f_r \geq f_0/2$ and exactly preserve the initial cell moments.

The same $\varepsilon > 0$ can preserve admissibility for every small r . In a fixed neighborhood W of X_*^0 , every bump remains inside the same cell. Its mass and first moment there are independent of r and of the configuration. Consequently C_{f_r} is independent of r on W and is $O(\varepsilon)$ close to C_{f_0} in $C^1(W)$; the attracting minimum continues to a common X_*^ε . On a compact tube around the baseline trajectory through a finite time T , let $m_T > 0$ bound its cell masses below and let $D = \text{diam}(\Omega)$. Subtracting centroid ratios gives

$$\|C_{f_r} - C_{f_0}\|_{\text{b}} \leq \frac{2D\varepsilon}{(1 - \varepsilon)m_T},$$

using $\|f_r - f_0\|_{L^1} \leq 2\varepsilon$. Gronwall's inequality with the baseline field's Lipschitz constant keeps the perturbed solutions $O(\varepsilon)$

close through T , uniformly in r . Choose T so the baseline has entered a smaller attracting neighborhood in W , then choose fixed ε small enough that all perturbed trajectories enter its continued basin. They converge to X_*^ε .

For any random estimated density Q , its mean $\bar{q}$ is independent of r , and convexity gives $\mathbb{E} D_{\text{KL}}(f_r \| Q) \geq D_{\text{KL}}(f_r \| \bar{q})$. For $E_r = \bigcup_{i,j} B(a_{ij}, r)$, one has $f_r(E_r) \geq \varepsilon$ and $\bar{q}(E_r) \rightarrow 0$. Binary data processing yields

$$D_{\text{KL}}(f_r \| \bar{q}) \geq \varepsilon \log \frac{1}{\bar{q}(E_r)} - \log 2 \rightarrow \infty.$$

Thus every estimator has infinite worst-case risk on the stated class.

Different endpoints from the same snapshot. Take $\Omega = [0, 1]$, $n = 2$, and $X_0 = (1/5, 3/5)$. Choose a nonnegative $\psi \in C_c^\infty((0.45, 0.55))$, symmetric about $1/2$, with $\psi(1/2) > 0$. For small $\varepsilon > 0$, $f_{\pm} = 1 \pm \varepsilon \psi''$ are strictly positive normalized densities. The perturbation lies inside the initial right cell and has zero mass and first moment, so both fields give $C_0 = (1/5, 7/10)$ and $V_0 = (0, \gamma/10)$. Symmetry and integration by parts give $\int_0^{1/2} \psi'' = 0$ and $\int_0^{1/2} z \psi'' = -\psi(1/2)$. Hence their distinct equilibria are

$$X_{\pm}^* = (1/4 \mp 2\varepsilon \psi(1/2), 3/4 \pm 2\varepsilon \psi(1/2)).$$

Their midpoint is $1/2$, so these are exactly their cell centroids. On $U = [0.15, 0.35] \times [0.55, 0.85]$, the uniform centroid map $C(X) = (b/2, (1+b)/2)$, $b = (x_1 + x_2)/2$, maps strictly into U and has block-norm Lipschitz constant $1/2$. Small C^1 perturbations preserve invariance and a contraction constant below one. Variation of constants for $\dot{X} = \gamma(C_f(X) - X)$ then gives exponential convergence from $X_0 \in U$ to each map's unique fixed point. At the uniform equilibrium, the coverage Hessian has eigenvalues $1/2$ and 1 ; continuity preserves positive definiteness at both perturbed equilibria. This verifies the common invariant neighborhood and both admissible, distinct limiting minima. $\square$

APPENDIX B – PROOF OF THEOREM 2

Define $R_f(X) = X - C_f(X)$ and $A_* = \frac{\partial R_f}{\partial X} \Big|_{X=X_*}$. Write $M_* = M_f(X_*)$ and $H_* = \nabla^2 \mathcal{H}_f(X_*)$. Each limiting Voronoi cell has positive mass because f is strictly positive and the limiting sites are distinct and interior. Thus M_* is invertible. The identity $\nabla \mathcal{H}_f(X) = 2M_f(X)R_f(X)$ and stationarity at X_* therefore imply $R_f(X_*) = 0$.

To differentiate this identity at X_* , observe that $R_f(X_* + Z) = A_* Z + o(\|Z\|)$ and $M_f(X_* + Z) = M_* + o(1)$. Since $R_f(X_*) = 0$, it follows that $\nabla \mathcal{H}_f(X_* + Z) = 2M_* A_* Z + o(\|Z\|)$. Consequently, $H_* = 2M_* A_*$. By Assumption 1, H_* is positive definite, so $A_* = \frac{1}{2} M_*^{-1} H_*$ is invertible.

Set $a = \sup_{Z \neq 0} \|A_*^{-1} Z\|_{\text{b}} / \|Z\|_{\text{b}}$. This constant is positive and finite because A_*^{-1} is a linear map on a finite-dimensional space. For every Z , the inequality $\|Z\|_{\text{b}} = \|A_*^{-1} A_* Z\|_{\text{b}} \leq a \|A_* Z\|_{\text{b}}$ gives $\|A_* Z\|_{\text{b}} \geq a^{-1} \|Z\|_{\text{b}}$.

Choose $\kappa_1 > a$ and set $\varepsilon = a^{-1} - \kappa_1^{-1} > 0$. Continuity of the Jacobian of R_f allows us to choose a sufficiently small convex

neighborhood U of X_* such that $\|(\frac{\partial R_f}{\partial X}(X) - A_*)Z\|_b \leq \varepsilon\|Z\|_b$ for every $X \in U$ and every Z .

For $X \in U$, put $e = X - X_*$. Convexity ensures that the segment $X_* + se$, $0 \leq s \leq 1$, lies in U . The fundamental theorem of calculus gives $R_f(X) = A_*e + E(X)$, where

$$E(X) = \int_0^1 \left(\frac{\partial R_f}{\partial X}(X_* + se) - A_* \right) e ds.$$

The chosen derivative bound implies $\|E(X)\|_b \leq \int_0^1 \varepsilon\|e\|_b ds = \varepsilon\|e\|_b$. The reverse triangle inequality now yields

$$\begin{aligned} \|R_f(X)\|_b &\geq \|A_*e\|_b - \|E(X)\|_b \\ &\geq (a^{-1} - \varepsilon)\|e\|_b = \kappa_1^{-1}\|X - X_*\|_b. \end{aligned}$$

At the observed configuration, $V_0 = \gamma(C_f(X_0) - X_0) = -\gamma R_f(X_0)$. Substitution therefore gives $\|X_* - X_0\|_b \leq (\kappa_1/\gamma)\|V_0\|_b$, which proves the first bound.

For the second bound, use $C_0 = X_0 + V_0/\gamma$ to obtain

$$\begin{aligned} \|X_* - C_0\|_b &\leq \|X_* - X_0\|_b + \gamma^{-1}\|V_0\|_b \\ &\leq \frac{\kappa_1 + 1}{\gamma}\|V_0\|_b. \end{aligned}$$

Thus we may take $\kappa_2 = \kappa_1 + 1$. Finally, $X(t) \rightarrow X_*$ implies that $X(t_0) \in U$ for all sufficiently large t_0 , so both bounds apply at all sufficiently late times. $\square$

APPENDIX C – PROOF OF THEOREM 3

Let $J(X) = \frac{\partial F_f}{\partial X}(X)$, $J_* = J(X_*)$, and $M_* = M_f(X_*)$. Since $F_f = -\gamma R_f$, the identity established in the preceding proof gives $J_* = -\gamma A_* = -(\gamma/2)M_*^{-1}H_*$. Conjugating by $M_*^{1/2}$ yields

$$M_*^{1/2}J_*M_*^{-1/2} = -\frac{\gamma}{2}M_*^{-1/2}H_*M_*^{-1/2}.$$

The matrix on the right is symmetric and negative definite: for every nonzero Z , its quadratic form equals $-(\gamma/2)(M_*^{-1/2}Z)^\top H_*(M_*^{-1/2}Z) < 0$. Its eigenvalues are therefore strictly negative. Similarity preserves eigenvalues, so J_* is Hurwitz and invertible.

By continuity of J , all eigenvalues remain in the open left half-plane when X is sufficiently close to X_* . Choose a bounded convex neighborhood U whose closure lies inside this region and inside the neighborhood from Theorem 2. The inverse $J(X)^{-1}$ is continuous on the compact set $\bar{U}$, so $B = \max_{X \in \bar{U}} \|J(X)^{-1}\| < \infty$.

The local Lipschitz assumption in Theorem 3 allows us to shrink U , if necessary, so that a finite constant L satisfies $\|J(X) - J(Y)\| \leq L\|X - Y\|$ for all $X, Y \in U$. Convergence of the trajectory gives a time T such that $X_0 = X(t_0) \in U$ whenever $t_0 \geq T$. For such a time, write $J_0 = J(X_0)$ and $e = X_* - X_0$.

Because the segment $X_0 + se$ lies in U , the fundamental theorem of calculus gives $F_f(X_*) = F_f(X_0) + J_0e + r$, where

$$r = \int_0^1 (J(X_0 + se) - J_0)e ds.$$

Using $F_f(X_*) = 0$ and $F_f(X_0) = V_0$, we obtain $0 = V_0 + J_0e + r$. Moreover, the Lipschitz bound gives

$$\|r\| \leq \int_0^1 Ls\|e\|^2 ds = \frac{L}{2}\|e\|^2.$$

For the block maximum norm, $\|Z\|_b \leq \|Z\| \leq \sqrt{n}\|Z\|_b$. Combining these inequalities with Theorem 2 therefore yields

$$\begin{aligned} \|e\| &\leq \sqrt{n}\|e\|_b \\ &\leq \frac{\sqrt{n}\kappa_1}{\gamma}\|V_0\|_b \leq q\|V_0\|, \quad q = \frac{\sqrt{n}\kappa_1}{\gamma}. \end{aligned}$$

Since J_0 is invertible, define $Y_\infty = X_0 - J_0^{-1}V_0$. The relation $0 = V_0 + J_0e + r$ implies $X_* - Y_\infty = e + J_0^{-1}V_0 = -J_0^{-1}r$. Consequently,

$$\begin{aligned} \|X_* - Y_\infty\| &\leq B\|r\| \\ &\leq \frac{BL}{2}\|e\|^2 \leq \frac{BLq^2}{2}\|V_0\|^2. \end{aligned}$$

Taking $K = BLq^2/2$ proves the quadratic error bound.

The Hurwitz property also confirms the endpoint interpretation: the affine system $\dot{Y} = V_0 + J_0(Y - X_0)$, $Y(0) = X_0$, has solution $Y(\tau) = Y_\infty + \exp(J_0\tau)(X_0 - Y_\infty)$, which converges to Y_∞ as $\tau \rightarrow \infty$. If $V_0 = 0$, Theorem 2 gives $X_0 = X_*$, and hence $Y_\infty = X_*$ as well. $\square$

APPENDIX D – SIMULATION SETUP

A. The Eight-Peak Deployment

Thirty UAS follow (1) on $[0, 1]^2$ with $\gamma = 1$, starting around $(0.5, 0.5)$ with coordinate standard deviation 0.025 (seed 9027). The synthetic density has a 0.02 uniform background; its remaining mass is split among eight normalized Gaussian components of width 0.04. In order, the eight centers are

$$\begin{aligned} &(0.20, 0.22), (0.49, 0.18), (0.79, 0.23), (0.82, 0.51), \\ &(0.76, 0.80), (0.48, 0.83), (0.18, 0.76), (0.16, 0.46), \end{aligned}$$

with weights $(0.11, 0.14, 0.09, 0.16, 0.13, 0.12, 0.15, 0.10)$. The estimator receives exact positions and velocities, not the density or its parameters. It uses a 32×32 Gaussian grid, width 0.03, $\eta = 0.02$, and $\lambda = 10^{-7}$.

Probes use $h = 2 \cdot 10^{-5}$ and stability margin $\delta = 10^{-8}$. Both predictors compute the required Voronoi-cell integrals using order-16 numerical quadrature. For the full-simulation baseline, we use the DOP853 solver to advance the fitted Lloyd dynamics until every vehicle's speed falls below 10^{-10} . All seven simulation runs reached this stopping condition.

B. The A-Shaped Field

We next use 24 UAS, initialized uniformly at random in $[0.1, 0.2] \times [0, 1]$ (seed 19273). The demand is an A centered at $(0.75, 0.5)$ in the right half of the square. Its two legs join $(0.55, 0.08)$ and $(0.95, 0.08)$ to $(0.75, 0.92)$; a crossbar joins the legs at height 0.43. Uniform line mass on these strokes, weighted by stroke length, is smoothed by isotropic Gaussians of width 0.035, normalized on the domain, and mixed with a 0.02 uniform background. Smoothing produces small tails

beyond the strokes; the background retains coverage demand outside the A.

The true fixed gain is 0.8, hidden from the estimator, which starts at 1.2 with prespecified bounds $[0.3, 1.8]$. We use a 24×24 Gaussian basis of width 0.035, $\eta = 0.02$, and $\lambda = 10^{-7}$. The observed times are

$$\mathcal{T} = \{0, 0.05, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, \\ 6, 8, 12, 16, 24, 32, 48, 64, 96, 128, 192, 256, 384, \\ 512, 640, 768, 1024\}.$$

Every third observed snapshot (zero-based index 2 modulo 3) is held out during gain selection. Updates occur at 0.5, 1, 2, 4, 8, 16, 32, and the seven forecast times 64, 128, 256, 512, 640, 768, and 1024; the initial density uses the gain guess. At every update, the selected gain is followed by an all-history density refit, as detailed in Appendix E.

APPENDIX E – GAIN ESTIMATION

Unknown-gain profiling. The physical gain is constant but its estimate may change as observations accumulate. For candidate $a > 0$, fit the RBF weights on training snapshots $\mathcal{T}_t$ with centroids $X + V/a$. Denote the resulting density by $\hat{f}_{a,\mathcal{T}_t}$. Define $\hat{F}_{a,t}(X) = a(C_{\hat{f}_{a,\mathcal{T}_t}}(X) - X)$. For evaluated candidates $\mathcal{G}_t \subset [\gamma_{\min}, \gamma_{\max}]$, choose the gain on disjoint validation snapshots $\mathcal{V}_t$ by

$$\hat{\gamma}_t \in \arg \min_{a \in \mathcal{G}_t} \frac{1}{n|\mathcal{V}_t|} \sum_{s \in \mathcal{V}_t} \|\hat{F}_{a,t}(X^{(s)}) - V^{(s)}\|^2. \quad (\text{E.1})$$

Then refit at $\hat{\gamma}_t$ using all available snapshots. The selection loss is not an independent test-error estimate. Ground-truth gain, density KL, and endpoint error enter only evaluation. This procedure is empirical: joint gain/density identifiability and convergence of the estimates are not guaranteed.

Algorithm E.1 Gain profiling and density refitting

Require: History through t ; initial/previous gain; fixed basis, ridge, gain bounds, and validation-split rule.

- 1: Split the available history into $\mathcal{T}_t, \mathcal{V}_t$.
 - 2: **if** $\mathcal{V}_t$ is empty **then**
 - 3: Retain the initial gain guess.
 - 4: **else**
 - 5: Tighten the gain bounds by observed-cell feasibility.
 - 6: **for** each candidate gain in the bounded search **do**
 - 7: Rebuild training moments for that gain.
 - 8: Fit the RBF density by Section IV-A on $\mathcal{T}_t$.
 - 9: Evaluate the held-out loss in (E.1).
 - 10: **end for**
 - 11: Select the gain with the smallest evaluated loss.
 - 12: **end if**
 - 13: Refit on all available snapshots at the selected gain.
 - 14: Return the gain and density for main-paper Algorithm 2.
-

On the unit square, cache

$$M_{rj} = \frac{1-\eta}{q_r} \int_{V_r} \phi_j, \\ P_{(r,d),j} = \frac{1-\eta}{q_r} \int_{V_r} (z_d - x_{r,d}) \phi_j, \\ p_{r,d} = \frac{\eta}{q_r} \int_{V_r} (z_d - x_{r,d}).$$

For candidate gain $a > 0$, the moment matrix, background offset, and fitted velocity become

$$A_{(r,d),j}(a) = P_{(r,d),j} - \frac{v_{r,d}}{a} M_{rj}, \\ b_{r,d}(a) = p_{r,d} - \frac{\eta v_{r,d}}{a}, \\ \hat{v}_{r,d}(a, w) = a \frac{(Pw)_{r,d} + p_{r,d}}{(Mw)_r + \eta}. \quad (\text{E.2})$$

Thus every historical row changes with the gain, without repeating quadrature. For a cell face $n^\top z \leq b$, centroid feasibility imposes $a \geq n^\top v_r / (b - n^\top x_r)$ whenever $n^\top v_r > 0$. The maximum of these bounds over the observed history, with relative margin 10^{-8} and lower limit 0.3, defines the search interval up to 1.8. Eleven equally spaced candidates and the previous gain are evaluated; the best neighboring interval is refined to requested tolerance $2 \cdot 10^{-4}$, with at most 18 scalar iterations. The best evaluated candidate is selected. OSQP uses scaled absolute/relative tolerances $2 \cdot 10^{-9}$ with at most 80 000 iterations. For the A experiment, validation uses every third snapshot (index 2 modulo 3); the initial gain is retained before validation exists.

Numerical settings. Basis parameters, observation times, and gain updates are specified in the main paper. Density KL uses a separate 240×240 grid. Both experiments use centered steps $h = 2 \cdot 10^{-5}$, stability margin $\delta = 10^{-8}$, and DOP853 tolerances $2 \cdot 10^{-9}$ (relative), $2 \cdot 10^{-11}$ (absolute), with maximum step 5. Eight-peak moment and prediction quadrature orders are 40 and 16; the A-shaped orders are 32 and 20. The A trajectory uses order 24 and its recorded velocities order 40. Full predictions stop at maximum speed 10^{-10} , with horizon caps 4096 and 16384, respectively; all seven forecasts in each experiment reach that threshold. The A reference endpoint is separately refined to speed below 10^{-13} . Fitted density and gain remain fixed throughout each forecast.

Stability and refinement. At all seven forecast times 64, 128, 256, 512, 640, 768, 1024, both true and fitted Jacobians are checked. In both experiments the first three are unstable and the last four Hurwitz, so reuse starts at 512. No eigenvalues are modified. Increasing eight-peak quadrature from 16 to 24 and halving the difference step preserves these signs and changes correction endpoints by at most $3.59 \cdot 10^{-5}$ RMSE; full-simulation endpoint refinement changes them by at most $6.78 \cdot 10^{-9}$. For the A experiment, order-32 Jacobians with half steps preserve all stability signs, with maximum relative operator-norm change $4.85 \cdot 10^{-8}$ and correction change $3.17 \cdot 10^{-7}$. Order-32 learned-endpoint refinement changes

positions by at most $3.20 \cdot 10^{-9}$. True-trajectory refinement changes observed positions by at most $2.14 \cdot 10^{-6}$ RMSE, and order-48 true-endpoint refinement changes the reference by $2.95 \cdot 10^{-14}$. At $t = 512$, the A full-simulation error remains 0.0857 after optimal matching, so it is not solely a label permutation.

Gain control and data checks. At $t = 1024$, A-field KL errors are 0.9954 with the initial gain fixed at 1.2, 0.0068 with estimated gain, and 0.0082 with the known true gain 0.8. Each control refits to the same history; neither control is used in selection. Moment fitting does not minimize true-density KL, so the ordering does not imply more accurate gain recovery. Checks verify normalized densities, consistent quadrature and cached velocities, recovery of a realizable RBF mixture, and causal data use with disjoint training/validation sets before the final all-history refit.

APPENDIX F – A THREE DIMENSIONAL EXAMPLE

We consider 24 UAS in $\Omega = [0, 1]^3$ following $\dot{X} = \gamma(C_f(X) - X)$, where $C_f(X)$ stacks the centroids of the three-dimensional Voronoi cells weighted by the demand density f . Initial horizontal positions are uniform in $[0.1, 0.2] \times [0.02, 0.98]$, and initial altitudes are independently uniform in $[0.15, 0.85]$ (seed 19273).

a) Density construction. The demand follows an A-shaped ridge when viewed from above. Its legs join $(0.55, 0.08, 0.8)$ and $(0.95, 0.08, 0.8)$ to $(0.75, 0.92, 0)$; a cross-bar connects the legs at $y = 0.43$. The ridge altitude is $h(y) = 0.8(0.92 - y)/0.84$, increasing from zero at the apex to 0.8 at the leg bottoms. For each of the three lifted strokes, write $s_j(u) = (1 - u)a_j + ub_j$, $0 \leq u \leq 1$, and let π_j be its projected xy length divided by the total projected stroke length. Using isotropic Gaussians $\phi_\sigma(\cdot; \mu)$ individually normalized on Ω , define

$$f(q) = 0.02 + 0.98 \sum_{j=1}^3 \pi_j \int_0^1 \phi_{0.055}(q; s_j(u)) du. \quad (\text{F.1})$$

This is a smooth, strictly positive probability density. The zero altitude specifies the ridge apex; Gaussian smoothing spreads demand above it. Integrating over altitude retains the A-shaped projection.

b) Estimation setup. The true gain $\gamma = 0.8$ is hidden. Cumulative motion observations are used to fit a positive Gaussian mixture and profile the gain over $[0.3, 1.8]$, starting from 1.2. Every third snapshot is held out for gain selection, followed by a density refit using all available snapshots. The density basis has $p = 12^3 = 1728$ centers on the full-cube midpoint grid $\{(i + 1/2)/12 : i = 0, \dots, 11\}^3$, spacing $1/12$, bandwidth 0.055, uniform background 0.02, and ridge coefficient 10^{-7} . This grid balances resolution and repeated-fitting cost; the bandwidth supplies the benchmark's fixed smoothness prior. The stroke geometry and limiting deployment are not supplied to the estimator. We use 29 snapshots through $t = 1024$ and report predictions at $t_0 = 0, 0.5, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 640, 768, 1024$.

c) Deployment prediction. For a prediction $\tilde{X}$, endpoint error is $[24^{-1} \sum_i \|\tilde{x}_i - x_{i,*}\|_2^2]^{1/2}$, using the true limiting positions $x_{i,*}$ and fixed vehicle labels. Figure 7 compares current positions, motion-inferred centroids $X_0 + V_0/\hat{\gamma}_{t_0}$, full simulation under the fitted field, and Jacobian corrections $X_0 - \hat{J}^{-1}V_0$ with a refreshed or reused matrix. The reused matrix, including its gain, is frozen at $t_0 = 256$. The refreshed matrix is stable at 256, 512, 640, 768, but unstable at all earlier reported times and at 1024; open markers indicate algebraic corrections without an attracting affine-model interpretation.

At $t = 512$, current-position and estimated-centroid errors are $3.78 \cdot 10^{-7}$ and $3.64 \cdot 10^{-7}$, compared with $1.99 \cdot 10^{-7}$ for the refreshed correction, $1.44 \cdot 10^{-7}$ for the reused correction, and 0.164 for full learned-field simulation. The latter converges to a stable equilibrium of the fitted field, which need not equal the true deployment. The estimated gain is 0.7997 at $t = 0.5$ and 0.8008 at 512.

d) Density recovery and numerical checks. Figure 6 recovers the projected A shape and its altitude gradient. Figure 8 shows KL falling from 2.967 initially to 0.0614 at $t = 8$ and 0.0530 at 1024. It need not decrease monotonically because the fitting objective uses motion constraints rather than KL. Density error can persist while position and velocity-anchored endpoint errors vanish. KL uses 64^3 Gauss–Legendre nodes; refinement to 80^3 at $t = 0, 512, 1024$ changes it by at most $5.4 \cdot 10^{-8}$. Full forecasts stop at maximum fitted speed 10^{-7} . Refining the seven late forecast equilibria changes them by at most $5.69 \cdot 10^{-6}$ RMSE. A refined true-trajectory calculation changes positions at 1024 by $2.60 \cdot 10^{-14}$ RMSE, so endpoint errors of order 10^{-13} at that time are near the numerical resolution.

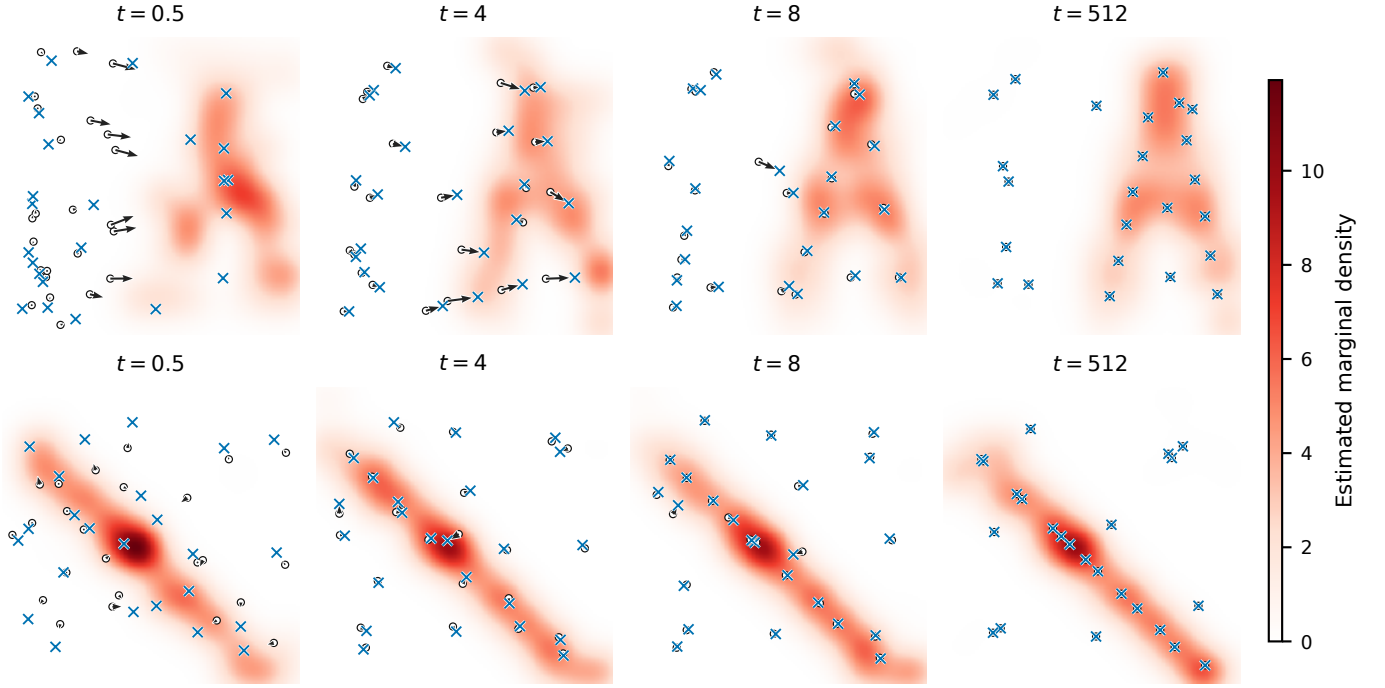


Fig. 6. Estimated 3D demand at $t = 0.5, 4, 8, 512$. Top: xy views with marginal density $\int_0^1 \hat{f}(x, y, z) dz$. Bottom: simultaneous yz views with marginal density $\int_0^1 \hat{f}(x, y, z) dx$ (y horizontal, z vertical). White circles show observed positions; blue crosses show centroid estimates $X + V/\hat{\gamma}_t$. Dark arrows show measured velocities, scaled by 0.25, 1, 1, 10^4 , respectively, in both rows. Centroid estimates are unscaled. All panels share the color scale.

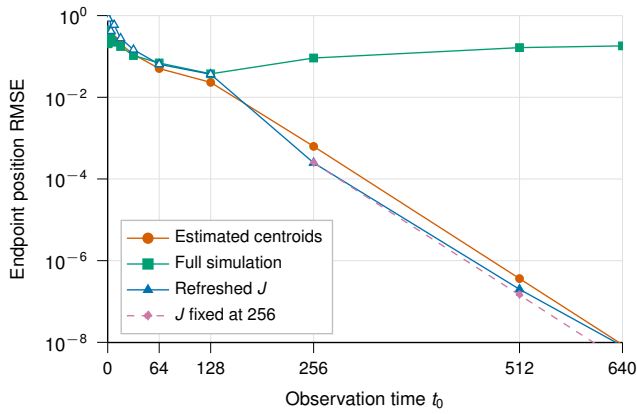


Fig. 7. Endpoint errors for the 3D example.

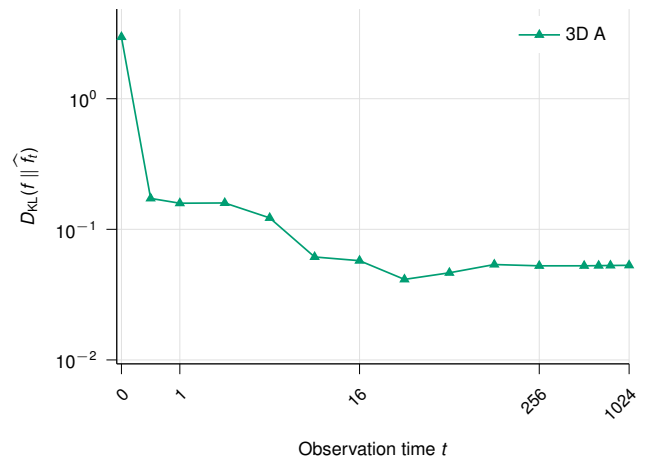


Fig. 8. Forward KL divergence $D_{\text{KL}}(f \| \hat{f}_t)$ for the fitted 3D density.